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Engineered base editors with reduced bystander editing through directed evolution

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Ramiro M. Perrotta[1,4], Svenja Vinke[1,4], Raphaël Ferreira[1,2,4], Michaël Moret[1], Ahmed Mahas[1], Anush Chiappino-Pepe[1,3], Lisa M. Riedmayr[1], Anna-Thérèse Mehra[1], Louisa S. Lehmann[1] & George M. Church[1,3]

Base editors enable precise genome modification but are constrained by bystander edits that limit their applicability. Existing strategies to enhance precision often compromise efficiency and remain highly sequence dependent. Here we present a parallel engineering approach that optimizes both guide RNAs and the deaminase enzyme to minimize bystander editing without sacrificing activity. We designed a library of 3′-extended guide RNAs and identified context-dependent variants that improved specificity. Using a precision-driven phage-assisted evolution system and protein language models, we evolved adenine base editor variants two- to threefold more precise than adenine base editor ABE8e while maintaining high efficiency across a library of thousands of human pathogenic contexts in vitro. Our findings establish a scalable framework for precision engineering of base editors, addressing a major challenge in genome editing.

Base editing has expanded the genome editing toolkit by offering high editing efficiencies, both in vivo and in vitro, without inducing double-strand breaks[1,2]. Adenine and cytosine base editors (ABEs and CBEs) catalyze A•T-to-G•C and C•G-to-T•A transitions within a five-nucleotide editing window[2,3]. When paired with a nCas9-SpRY variant[4], these enzymes can induce all four transition mutations and correct most known human pathogenic single nucleotide polymorphisms (SNPs)[1], overcoming the constraints imposed by protospacer adjacent motif (PAM) sequence requirements. However, the promiscuous activity of fused deaminases allows the conversion of not only the target nucleotide, but also nontarget bases within the editing window. This bystander effect has the potential to result in missense mutations, presenting limitations for the translation into clinical applications[1].

Despite efforts to minimize bystander edits, they often prove unavoidable, resulting in coding nonsilent mutations and mis-splicing variants within the target gene. For ABEs, assuming the presence of at least one bystander adenine within the activity window, there is a substantial 38% probability that bystander edits will lead to nonsilent mutations, potentially with detrimental biological effects[1]. These findings raise profound concerns about the clinical feasibility and safety of base editing technologies[5].

Recent strategies have been used to decrease bystander activity. For example, through rational design of deaminases with narrowed editing windows[6–8], directed evolution[9,10], modified gRNAs[11] and the use of gRNAs in combination with SpRY base editors (BEs) that accommodate the editing window to avoid nonsynonymous mutations[12]. Current evolved BEs featuring restricted editing windows exhibit diminished editing activity[10,13]. The BE efficiency and editing pattern are highly influenced by the complex interaction between BEs, guide RNAs (gRNAs) and target sequences[14]. In addition to the constraints posed by PAM requirements and limited activity windows, BEs exhibit sequence context preferences that substantially impact editing efficiency[14–16].

Unlocking the full potential of base editing for therapeutic applications requires the development of a robust and adaptable methodology for designing new base editing strategies that are both efficient and precise.

[1]Department of Genetics, Harvard Medical School, Boston, MA, USA. [2]Department of Health Technology, Technical University of Denmark (DTU), Kongens Lyngby, Denmark. [3]Wyss Institute for Biologically Inspired Engineering, Harvard University, Boston, MA, USA. [4]These authors contributed equally: Ramiro M. Perrotta, Svenja Vinke, Raphaël Ferreira. e-mail: Ramiro_perrotta@hms.harvard.edu; gchurch@genetics.med.harvard.edu

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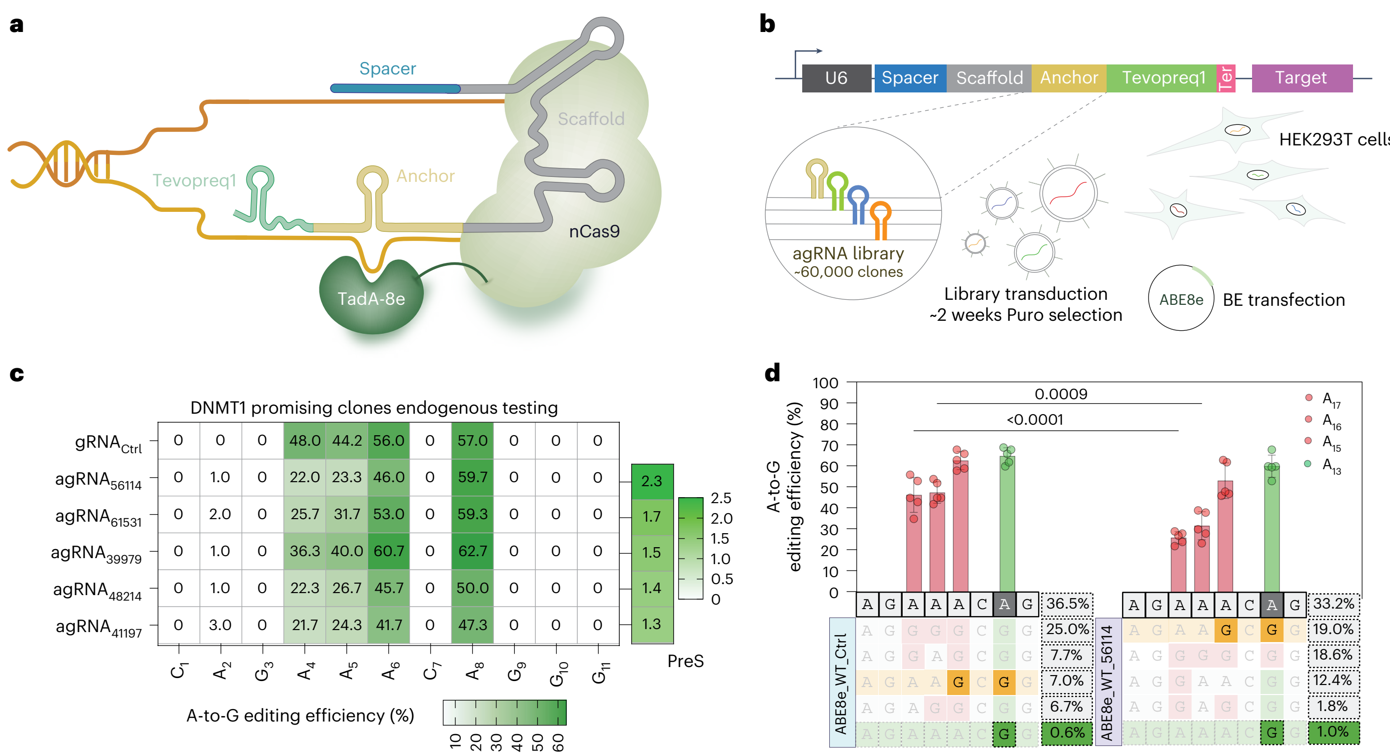


Fig. 1 | Impact of 3′-extended gRNAs on base editing outcomes at the human DNMT1 locus. a, Schematic representation of the agRNA (yellow) integrated into the base editing system. **b**, Overview of the experimental workflow and agRNA library design. More than 60,000 different agRNAs were cloned downstream of the scaffold within a 300-nt DNA cassette that also includes the target DNA site. **c**, Editing pattern and PreS scores of the five top-performing agRNAs at the endogenous DNMT1 locus using plasmid-based delivery in HEK293T cells (mean of $n = 3$ independent replicates). **d**, Quantification of editing efficiency by Sanger sequencing (barplots; mean ± s.d. of $n = 5$ independent replicates) and by NGS (abundance plots). NGS plots show the five most frequent edited alleles, with the green-highlighted row indicating the desired allele harboring only the intended A-to-G edit without bystander substitutions (mean of $n = 3$ independent replicates). P values were calculated by two-way ANOVA followed by Šídák's multiple comparisons test. Abundance plots in **d** created with BioRender.com.

This work presents a multifaceted approach to designing a base editing system that minimizes bystander editing while maintaining high efficiency in vitro. Our strategy integrates three complementary techniques: (1) gRNA engineering to reduce bystander editing, (2) phage-assisted noncontinuous evolution (PANCE[17]) to selectively evolve BEs with improved precision, and (3) protein language models (PLMs) to rationally engineer deaminases with optimized activity.

For gRNA engineering, we designed and tested a library of ~60,000 different 3′-extended sgRNAs, or anchor-guide RNAs (agRNAs), to improve the precision of ABEs. The most promising agRNA candidate from this library screening was then used as part of a PANCE system to evolve a more precise TadA-8e enzyme. Using a dual selection pressure (favoring precise editing at the target site while penalizing bystander edits) we identified several variants with narrowed editing windows. Notably, our PANCE-evolved V28C variant exhibited enhanced on-target efficiency while significantly reducing bystander editing. Editing patterns across ~12,000 pathogenic mutations demonstrated that V28C is around two- to threefold more precise and ~20% more efficient than ABE8e in the tested sites.

Recognizing the potential for further optimization, we also explored a complementary computational approach leveraging recent advances in deep learning for protein engineering. We aimed to design variants with improved fitness in a single step without requiring extensive experimental screening. We hypothesized that integrating two orthogonal methods (experimental evolution and machine learning (ML)) could reveal mutually beneficial substitutions while mitigating these inefficiencies. In line with this, we applied ML to computationally design TadA-8e mutants with improved precision and efficiency. Among these, the M151E mutation significantly narrowed the editing window while increasing on-target editing.

By integrating gRNA engineering, directed evolution and ML, this work provides a systematic strategy to enhance base editing precision without compromising efficiency. These advancements offer a path toward safer and more effective therapeutic applications.

Results

3′ gRNA extensions narrow the editing window and reduce bystander editing

The mechanism underlying the adenine deamination process involves that the TadA-8e domain engages with the exposed single-stranded region of the PAM-distal nontarget strand[18] (Supplementary Fig. 1a). The TadA-8e deaminase, when attached to the nCas9 protein, induces specific editing patterns within narrow DNA regions, covering several base pairs. This connection limits the enzyme to act on certain nucleotides, defining the editing window. As different DNA contexts are relatively diverse substrates that require the enzyme to accept structure variations within its active site, the DNA strand in the active site has a certain degree of freedom to move in this position.

Our aim was to establish a system that stabilizes the DNA strand within the active site. This restricted movement may result in a smaller editing window, thus minimizing the bystander effect. Therefore, we decided to explore the possibility of adding nucleotides to the 3′ end of the gRNA scaffold. These agRNAs were designed as a library of short sequences and entire hairpin structures (counter-loops) at the opposite side of the editing loop to introduce structures that may sterically restrict the movement of the DNA and TadA enzyme even further (Fig. 1a and Supplementary Fig. 1b). This design process yielded an agRNAs library containing 60,000 candidates. To screen the library in a high-throughput manner, we constructed a lentiviral vector with the editing target downstream of the agRNA. With the agRNA and the target being in a 300-nucleotide window, it is possible to perform next-generation sequencing (NGS) on the sequence in this region after the editing process and to analyze the editing pattern for each library clone (Fig. 1b).

The tested library was designed to target a site in the human *DNMT1* locus, being an optimal candidate for screening, both for high accessibility for editing in HEK293T cells, and an adenine rich context within the editing window. The ABE8e-nSpCas9-WT BE in combination with a non-3′extended guide ($sgRNA_{Ctrl}$) showed a high editing efficiency for the four adenines in the editing window (A_4, A_5, A_6 and A_8). As we detected a slight preference towards the two adenines at position 6 and 8 with almost the same editing efficiency, we decided to design the library to precisely edit A_8. With the current editing tools, undesired bystander edits in the selected context remain unavoidable.

We first selected anchors that showed higher efficiency and lower bystander editing in the *DNMT1* agRNA library (Supplementary Fig. 1c). We selected the five best candidates to test in the native context in HEK293T cells, and all of them showed a decrease in bystander editing (Fig. 1c). We introduced a precision score (PreS = (percentage on-target edit/average bystander editing) × (percentage on-target edit variant/percentage on-target edit ABE8e)), to select those clones that showed higher on-target editing and reduced bystander editing. Clone 56114 ($agRNA_{56114}$) showed higher PreS in A_8, with a significant editing reduction of 44% in A_{17} and 34% in A_{16} (Fig. 1c,d and Supplementary Fig. 1d). NGS abundance analysis showed an increase of 2.4-fold (±0.2) in A_8/A_6 editing and 1.5-fold (±0.18) in A_8 (Supplementary Fig. 1e). To test the reproducibility of this effect in different cell lines, we also tested the $agRNA_{56114}$ in HeLa and HepG2 cells, obtaining similar bystander reduction patterns (Supplementary Fig. 1f).

Precision-driven selection circuit enables the evolution of adenine base editors with narrowed editing windows

Although we observed a robust reduction in bystanders, we were far from generating a perfect edit in position A_8. To refine the editing, we decided to evolve the enzyme TadA-8e to further decrease bystander edits and to evaluate how the combination of new variants and agRNA could impact the editing pattern.

PANCE has been used in the past to increase the activity of ABEs[19]. Unlike rational design or saturation mutagenesis, phage-assisted evolution offers an unbiased evolution system that couples mutational diversification with a functional selection pressure[20]. Our goal was to design an alternative selection method that enables us to evolve variants with decreased bystander edits and that can work with agRNAs. Therefore, we needed a selection pressure that decreases the phage titer in response to bystander edits, but also increases it upon perfect editing, to prevent the evolution of an inactive TadA enzyme. To achieve this, we linked the activity of our BE encoded on the M13 phage genome with the expression of the gene *III* (encoding for the pIII protein) that is required for phage replication[20]. As the effect of pIII amino acid exchanges on phage infectivity is well characterized[21], we identified groups of two to three amino acids that drastically decrease phage replication. Toward this goal, we mutated the codons of selected amino acids in pIII to be T/A rich, which make these contexts responsive to A•T-to-G•C editing. We identified three contexts in which bystander edits would result in a decrease in phage titer, whereas the perfect edit generates an adaptive mutation selecting that also selects for targeted efficiency (Fig. 2a and Supplementary Fig. 2a). Each identified pIII variant was cloned into individual plasmids (SP1–3) (Fig. 2b). The accessory plasmid encoded the corresponding agRNA targeting the single nucleotide variant (SNV), as well as a C-terminal Intein-nSpCas9-SpRY. Once the selection phage carrying the N-terminal Intein–TadA-8e fusion protein infects the cell, a functional BE is expressed, capable of performing the editing. This way, our selection circuit enables the unbiased evolution of BEs toward both high precision and high efficiency, aiming to overcome the conventional trade-off between reducing bystander activity and maintaining editing efficiency.

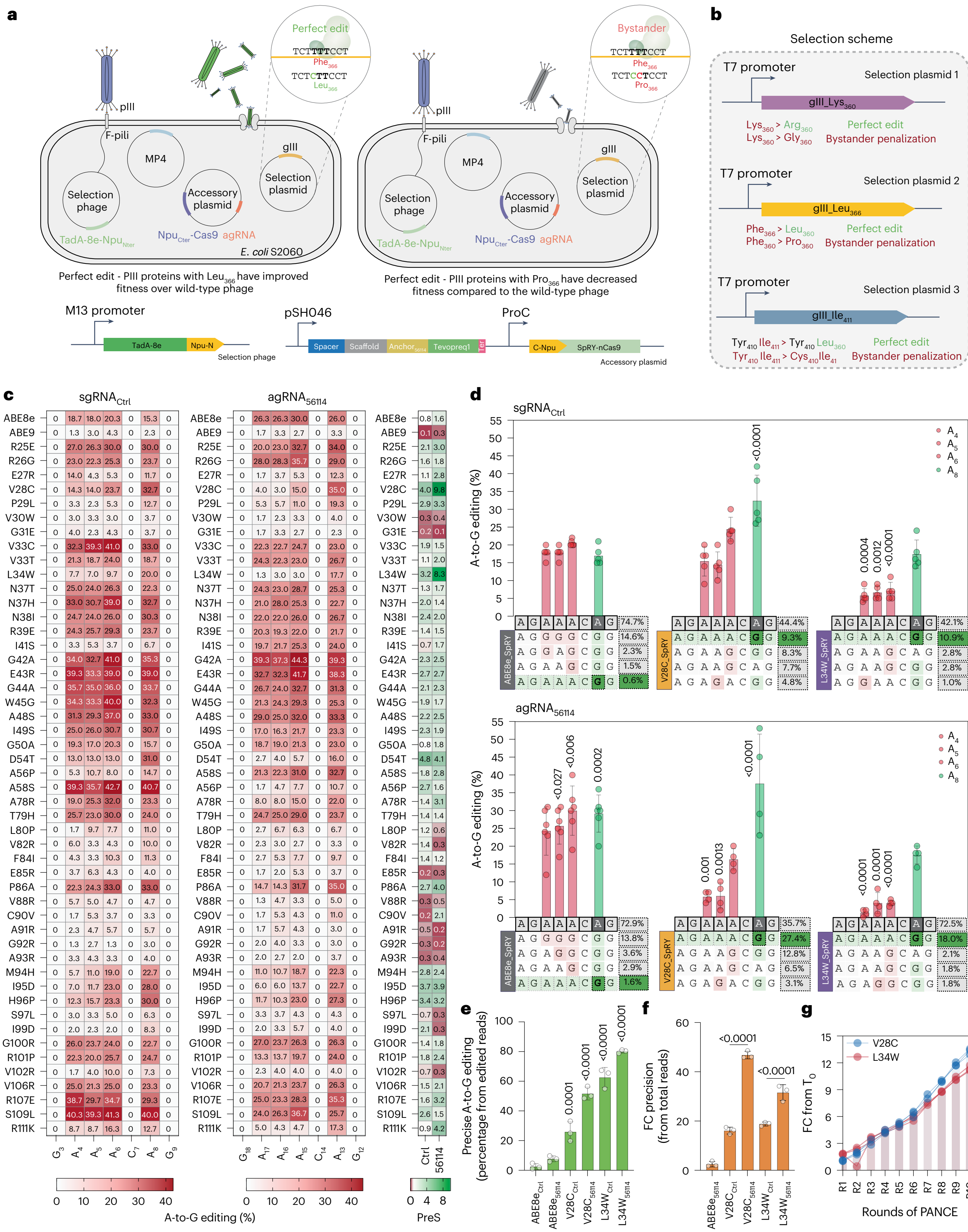
Phage DNA was isolated from each PANCE round and sequenced using NGS (Supplementary Fig. 2b). We observed an increase in phage titer after round four, suggesting an increased activity of the enzyme mutants encoded by the phage (Supplementary Fig. 2c).

The three replicates showed almost identical enrichment of the same variant over the course of the evolution (Spearman's correlation 0.95 ± 0.13 (mean ± s.d.)) (Supplementary Tables 1 and 2). The sequencing data also showed a slow enrichment of different mutants with amino acid exchanges, with a maximum enrichment of 3–4% of the new amino acid compared to the wild type (Supplementary Fig. 2d).

Having determined the TadA mutational landscape, we selected the top 50 most enriched mutations and individually tested the editing pattern in the human *DNMT1* locus in HEK293 cells (Fig. 2c). We detected variants with both higher efficiency and reduced bystander editing pattern at position A_8, when compared with the Abe8e-SpRY BE (Fig. 2c). We also benchmarked our PANCE-evolved TadA variants against the ABE9 BE (both WT and SpRY) that showed low editing efficiency and no relative bystander reduction in the *DNMT1* site (Fig. 2c). Variants displaying V28C and L34W showed the highest PreS with both $sgRNA_{Ctrl}$ and $agRNA_{56114}$ (Fig. 2b,c).

NGS abundance analysis demonstrates that ABE8e rarely achieved perfect editing (Fig. 2d). Compared to the ABE8e-WT, the SpRY variant exhibited a modest reduction in bystander editing with the $agRNA_{56114}$ while simultaneously enhancing overall editing efficiency (Fig. 2d). The primary outcome for both V28C and L34W was perfect editing, with an average of 9.3% (±1.13) and 10.9% (±0.63) respectively, when used with the $sgRNA_{Ctrl}$ (Fig. 2d). V28C, in combination with $agRNA_{56114}$, not only reduced bystander editing but also increased perfect editing to 27.4% (±1.14). L34W increased to 18% (±2.5) the on-target editing and kept all bystander editing below 5% (Fig. 2d). Of the reads

Fig. 2 | PANCE of precise ABEs. a, Schematic of the PANCE selection circuit. The selection phage encoded the TadA-8e adenine deaminase fused to a C-terminal intein, and the selection plasmids (SP1–3) encode the spCas9-SpRY fused to a N-terminal intein. **b**, Each selection plasmid encodes a different pIII sequence carrying a SNV together with the corresponding agRNA to correct the SNV to the wild-type sequence. Once the selection phage and the selection plasmid meet in the same cell, precise editing restores the wild-type pIII sequence, enhancing phage infectivity. In contrast, bystander edits generate amino acid substitutions that reduce phage replication. **c**, Editing pattern of top enriched ABE-SpRY variants isolated during the PANCE experiment. PreS reflects on-target editing versus bystander editing efficiency (mean of $n = 3$ independent replicates). **d**, Comparative editing analysis of ABE8e-SpRY and the two top PANCE-derived variants, V28C and L34W, at the endogenous human *DNMT1* locus with or without the $agRNA_{56114}$. Editing efficiencies were quantified by Sanger sequencing (barplots, mean ± s.d., $n = 5$ independent replicates except ABE8e-SpRY $agRNA_{56114}$, $n = 6$, and L34W-SpRY $agRNA_{56114}$, $n = 4$) and by NGS (abundance plots). Abundance plots display the four most frequent edited alleles, with the green-highlighted row indicating the desired allele containing only the intended A-to-G edit and no bystander substitutions (mean of $n = 3$ independent replicates). *P* values were determined using two-way ANOVA followed by Dunnett's multiple comparisons test (relative to ABE8e-SpRY $sgRNA_{Ctrl}$). **e**, Precise A-to-G editing at position A8 within edited reads (mean ± s.d.; $n = 3$ independent replicates). *P* values were determined using one-way ANOVA followed by Dunnett's multiple comparisons test (versus ABE8e-$SpRY_{Ctrl}$). **f**, FC in editing precision relative to ABE8e-SpRY $sgRNA_{Ctrl}$ calculated from total reads (mean ± s.d.; $n = 3$ independent replicates). *P* values were determined using one-way ANOVA followed by Tukey's multiple comparisons test. **g**, FC of the ratio of the V28C and L34W mutation across ten rounds of evolution ($n = 3$). Abundance plots in **d** created with BioRender.com.


a
Perfect edit
TCTTTTCCT
Phe366
TCTCTTCCT
Leu366
pIII
F-pili
MP4
gIII
Selection plasmid
Selection phage
Accessory plasmid
TadA-8e-NpuNter
NpuCter-Cas9 agRNA
E. coli S2060
Perfect edit - PIII proteins with Leu366 have improved fitness over wild-type phage
Bystander
TCTCCTCCT
Pro366
Perfect edit - PIII proteins with Pro366 have decreased fitness compared to the wild-type phage
M13 promoter
TadA-8e
Npu-N
Selection phage
pSH046
Spacer
Scaffold
Anchor56114
Tevopreq1
Ter
ProC
C-Npu
SpRY-nCas9
Accessory plasmid
b
Selection scheme
T7 promoter
Selection plasmid 1
gIII_Lys360
Lys360 > Arg360 Perfect edit
Lys360 > Gly360 Bystander penalization
Selection plasmid 2
gIII_Leu366
Phe366 > Leu360 Perfect edit
Phe360 > Pro360 Bystander penalization
Selection plasmid 3
gIII_Ile411
Tyr410 Ile411 > Tyr410 Leu360 Perfect edit
Tyr410 Ile411 > Cys410 Ile41 Bystander penalization
c
sgRNACtrl
agRNA56114
A-to-G editing (%)
PreS
d
sgRNACtrl
agRNA56114
A-to-G editing (%)
ABE8e_SpRY
V28C_SpRY
L34W_SpRY
e
Precise A-to-G editing (percentage from edited reads)
f
FC precision (from total reads)
g
V28C
L34W
FC from T0
Rounds of PANCE

for L34W in combination with agRNA$_{56114}$, 80.8% (±0.5) were perfect editing (Fig. 2e). V28C agRNA$_{56114}$ showed 53.4% (±4.6) of reads with perfect edits and the highest fold-change (FC) improvement when compared with ABE8e sgRNA$_{Ctrl}$ (47.24 ± 1.95) (Fig. 2f). Both variants exhibited enrichment across all rounds of PANCE evolution (Fig. 2g). nSpCas9-WT ABEs with the new mutations also showed increased precision (Supplementary Fig. 3a,b).

We did not observe an increase of indel byproducts (Supplementary Fig. 3g) and DNA Cas9-dependent off-target editing of ABE8e-SpRY and our variants, across four different sites (Supplementary Fig. 3c). Next, we performed an orthogonal R-loop assay[22] to access the Cas9-independent on-target editing, observing a substantial decrease in both variants across five different sites (Supplementary Fig. 3d). Finally, we analyzed RNA off-target editing by RNA sequencing. Both evolved enzymes significantly reduced the A-to-I deamination (Supplementary Fig. 3e). Cells edited with ABE8e-V28C-SpRY experienced ~7.9-fold less A-to-I deamination than ABE8e-SpRY (Supplementary Fig. 3f). This reduction was even higher for ABE8e-L34W-SpRY treated cells with a reduction of ~20.4-fold (Supplementary Fig. 3f).

Our evolution strategy successfully yielded TadA variants with significantly enhanced precision, as demonstrated at the *DNMT1* site, where the combination with agRNA$_{56114}$ achieved striking accuracy. The enriched variants, V28C and L34W, exhibited substantial reductions in bystander editing while maintaining or improving on-target efficiency. We also observed a significant reduction in Cas9/gRNA independent off-target deamination on DNA and RNA.

However, the fast pace of the selection of the phages might produce enzymes with the desired function but lacking stability. To accelerate and expand the search for optimal BE variants, integrating PLMs could provide a complementary strategy, enabling the in silico prediction of beneficial mutations before experimental validation.

ML-guided ABE8e design unveils new mutations that increase precision

Improving protein fitness traditionally involves screening a vast library of random mutations and selecting those that enhance a desired function under specific pressures (directed evolution). However, our PANCE experiment demonstrates that, even with increased fitness, specific mutational patterns across protein families are essential for evolvability. In general, most random mutations will be destabilizing or neutral for protein function (Fig. 3a).

In contrast, PLMs trained on massive, nonredundant protein sequence datasets can learn these general, evolutionarily plausible mutational patterns[23,24]. This knowledge can be leveraged to predict mutations likely to be beneficial, guiding protein evolution more efficiently. Following training, these models can be used to predict the probability distribution of each amino acid at any given position along a protein sequence, where the probability distribution reflects the knowledge acquired by the models on their training dataset. Positions where the model assigns a higher probability to an amino acid than the wild-type residue are considered more likely than a random pick to yield a positive effect on the protein fitness. This offers a promising and potentially more efficient alternative to traditional directed evolution methods.

In line with this, we used an ensemble of PLMs[23] to predict mutations likely to yield a positive effect on the TadA-8e sequence, as combining predictions from several models has been shown to increase prediction quality, notably on the task of increasing protein fitness[23,24]. We identified 21 evolutionary plausible mutations that we tested individually for their editing pattern in the *DNMT1* site (Fig. 3a). We observed a different editing pattern, favoring A_6, in contrast to our PANCE-evolved variants where A_8 showed improved efficiency (Fig. 3b). M151E showed the higher PreS towards A_6 with both sgRNA$_{Ctrl}$ and agRNA$_{56114}$. Of the 21 predicted mutations, 11 reverted to the amino acid found in the ancestral TadA (Fig. 3c). Ten of these reversions (A48P, F84L, V106A, R111T, Y123H, C146S, P152R, Y149F, V155E and N157K) showed increased PreS, while N108D abolished ABE8e editing at the *DNMT1* site (Fig. 3c). These data suggest a possible potential functional redundancy in some mutations present in the ABE8e enzyme.

NGS abundance analysis of M151E editing showed increased precision in position A_6, with the highest FC in combination with the and agRNA$_{56114}$ (25.7-fold ±2.4) and a 15.4-fold (±1.3) in A_8 (Fig. 3d,e).

Next, we ML-evolved our best PANCE-derived variant V28C (highest PreS) using the same approach. As expected, the model suggested to revert the 28C mutation as well as other ABE8e specific mutations towards the ancestral deaminase (Fig. 3f). We decided to combine the mutation with the highest PreS together with the V28C, to determine whether this approach can be additive. The V28C-M151E variant showed reduced editing efficiency at *DNMT1* position, but precise edit in other contexts such as Site9 (Fig. 3g and Supplementary Fig. 4a,b). These data suggest that the addition of the M151 mutation can improve precision, but with the cost of reducing efficiency and context compatibility (similar to other enzymes designed by rational design[6,10,13]). Although variants such as V28C-M151E or ABE9 exhibit greater context dependency, they remain highly valuable because of their potential to minimize bystander editing in certain clinically relevant contexts.

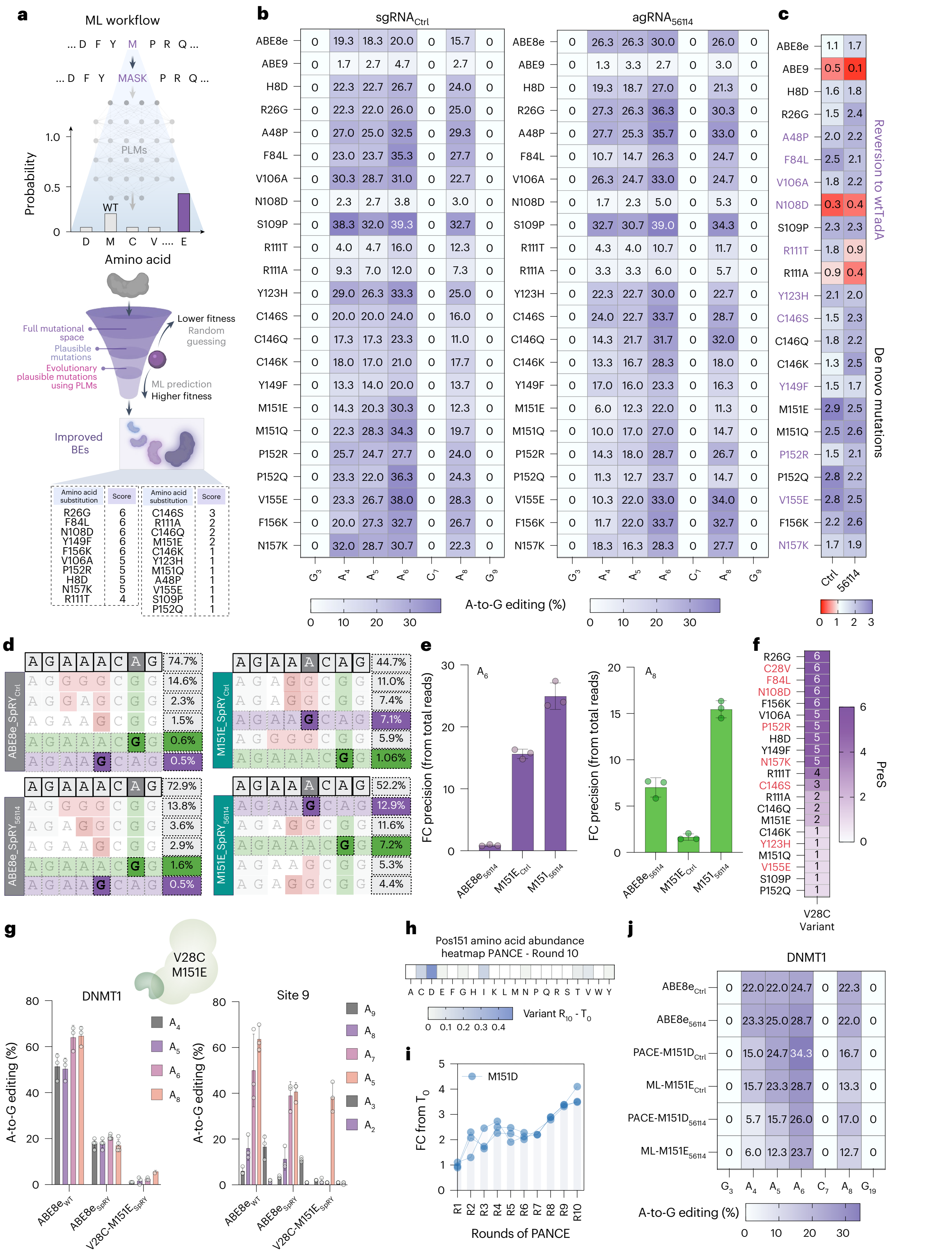
We decided to cross-reference the M151E mutation with the amino acid exchanges we observed during the PANCE. When we evaluated the amino acid substitutions at position 151, we observed an enrichment in aspartic acid across the different rounds of evolution (Fig. 3h,i). Both glutamic acid and aspartic acid are negatively charged amino acids, delivering similar editing patterns at the *DNMT1* site (Fig. 3j).

PLMs have proven valuable for enhancing protein stability and function but lack the ability to drive evolution toward a specific function. However, they can complement directed evolution approaches like PANCE by fine-tuning function through optimized protein stability. Our results demonstrate that integrating PLM-guided design with PANCE-driven selection enables the discovery of BE variants with both enhanced precision and efficiency.

ABE8e-V28C achieves superior precision and efficiency across diverse genomic sites

To further assess the performance of our variants across diverse genomic contexts, we analyzed editing patterns across a library of human pathogenic sites, including NGG and NG- PAMs[14] (Fig. 4d).

Fig. 3 | ML-guided design of ABE variants with reduced bystander editing. **a**, Illustrative output of PLM predictions for plausible substitutions across TadA-8e. The *y* axis reflects relative model scores (not normalized probabilities). **b**, Editing efficiencies at the human *DNMT1* for ABE ML-predicted single amino acid exchange mutants tested with sgRNA$_{Ctrl}$ and agRNA$_{56114}$ in HEK293T cells. **c**, PreS score of tested variants. Red highlighted residues denote reversions to the wild-type TadA amino acid. **d**, NGS abundance analysis of the control and the top ML-derived variant M151E-SpRY in the human *DNMT1* locus with and without agRNA$_{56114}$. Abundance plots display the five most frequent edited alleles, with the green-highlighted row indicating the bystander-less edit at position A_8 and in purple at position A_6 (mean of $n = 3$ independent replicates). **e**, FC of precise A-to-G edits at positions A_8 (left) or A_6 (right). **f**, Predicted mutations for the V28C variant. Numbers indicate how many of six independent models identified each substitution as a plausible mutation. Purple highlighted mutations are mutations that revert to the wild-type TadA amino acid at that position. **g**, Editing efficiencies at the endogenous *DNMT1* (top) and Site9 (bottom) loci for the double mutant V28C-M151E-SpRY. **h**, Amino acid frequencies at position 151 after the PANCE experiment (mean of $n = 3$ independent replicates). **i**, FC in the ratio of the M151D-SpRY mutation over successive PANCE rounds. **j**, Comparison of editing pattern at the *DNMT1* locus between position 151 variants generated by PANCE (M151D) or ML (M151E), quantified by Sanger (mean ± s.d.; $n = 3$ independent replicates). Abundance plots in **d** created with BioRender.com.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... D F Y MASK P R Q ...
PLMs
Probability
WT
Amino acid
Full mutational space
Plausible mutations
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Lower fitness
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Score
b
sgRNACtrl
agRNA56114
A-to-G editing (%)
c
Reversion to wtTadA
De novo mutations
Ctrl
56114
d
ABE8e_SpRYCtrl
M151E_SpRYCtrl
ABE8e_SpRY56114
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e
FC precision (from total reads)
ABE8e56114
M151ECtrl
M15156114
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PreS
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V28C M151E
DNMT1
Site 9
A-to-G editing (%)
ABE8eWT
ABE8eSpRY
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Variant R10 - T0
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M151D
FC from T0
Rounds of PANCE
j
DNMT1
ABE8eCtrl
ABE8e56114
PACE-M151DCtrl
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A-to-G editing (%)

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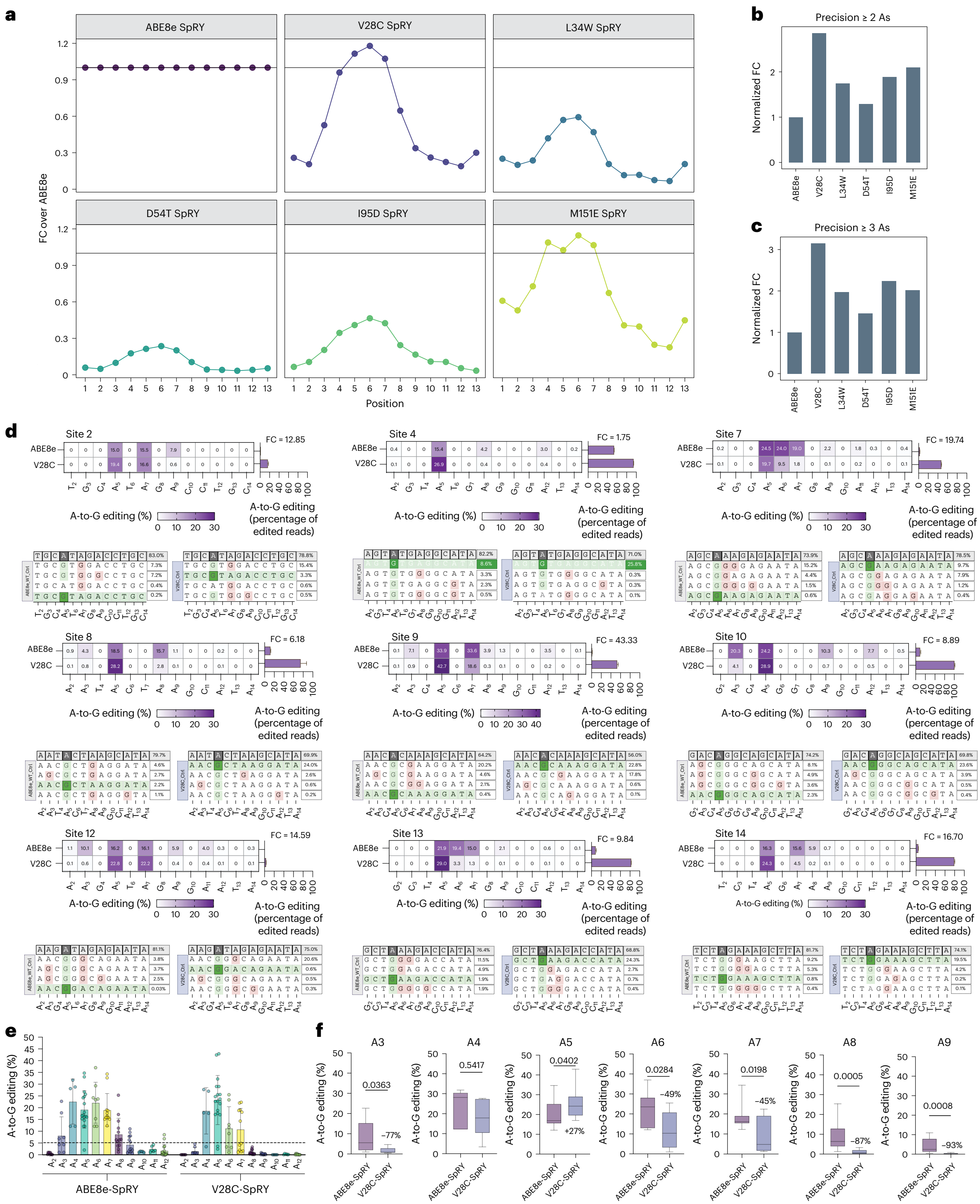

Fig. 4 | Genome-wide editing profiles of evolved ABEs. a, Normalized FC histograms of PANCE and ML-derived variants relative to ABE8e-SpRY across thousands of human pathogenic variants. **b**,**c**, FC in precision correction of pathogenic mutations at target sites containing more than two (**b**) or three (**c**) adenines. **d**, Editing pattern at 12 endogenous DNA sites edited with V28C or ABE8e_SpRY and measured by NGS. Abundance plots display the four most frequent edited alleles, with the green-highlighted row indicating the desired allele containing only the intended A-to-G edit and no bystander substitutions (mean of $n = 3$ independent replicates). **e**,**f**, Comparison of A-to-G editing efficiencies at positions A_3–A_9 between ABE8e-SpRY and V28C-SpRY across 12 human genomic loci. Each point represents the mean editing efficiency from $n = 3$ independent replicates containing an adenine at the indicated position. **f**, Statistical comparison of editing efficiencies for each adenine plotted in **e**. Boxplots display the median (center line), the 25th and 75th percentiles (bounds of the box) and whiskers extending to the minimum and maximum values. *P* values were determined using a two-tailed unpaired *t*-test. No adjustments for multiple comparisons were applied. Abundance plots in **d** created with BioRender.com.

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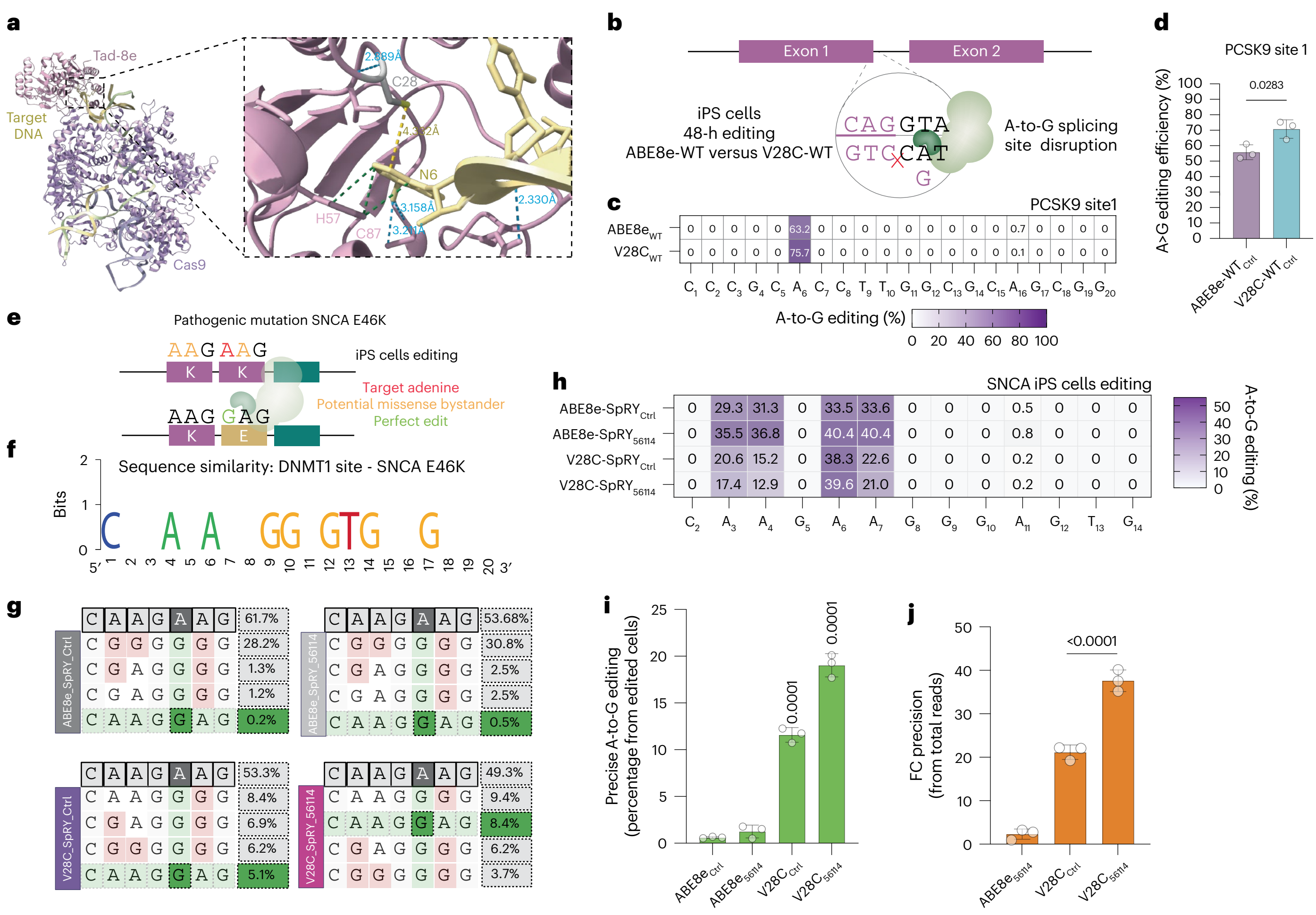

Fig. 5 | Performance of V28C as an editing tool for correcting human pathogenic mutations. a, In silico structural model showing the predicted effect of the V28C mutation on the interaction between TadA-8e and the target DNA. **b**, Schematic of the *PCSK9* editing strategy using ABE to disrupt a splicing site. **c**, Editing pattern of the *PCSK9* site correction by ABE8e-WT and V28C-WT variants (NGS, mean of $n = 3$ independent replicates). **d**, Quantification of A-to-G editing at the *PCSK9* target site measured by NGS (mean ± s.d., $n = 3$ independent replicates). *P* values were determined using a two-tailed unpaired *t*-test. **e**, Schematic of the SNCA E46K mutation. **f**, Sequence similarity between guides targeting the human *DNMT1* locus and the SNCA E46K mutation. **g**, NGS abundance plots showing the editing outcomes of the SNCA E46K mutation after treatment with ABE8e-SpRY or V28C-SpRY using agRNA$_{Ctrl}$ and agRNA$_{56114}$ (mean of $n = 3$ independent replicates). The sequence highlighted in green represents the perfect edit. **h**, Editing pattern of the SNCA E46K mutation correction by ABE8e-SpRY and the V28C-SpRY variant with agRNA$_{Ctrl}$ and agRNA$_{56114}$, (NGS, mean ± s.d.; $n = 3$ independent replicates). **i**, Quantification of precise A-to-G editing (A_6) within edited reads (mean ± s.d.; $n = 3$ independent replicates). *P* values were determined using one-way ANOVA followed by Dunnett's multiple comparisons test (versus ABE8e-SpRY$_{Ctrl}$). **j**, FC in editing precision relative to ABE8e-SpRY sgRNA$_{Ctrl}$ calculated from total reads (mean ± s.d.; $n = 3$ independent replicates). *P* values were determined using one-way ANOVA followed by Tukey's multiple comparisons test. Graphic in **f** created using weblogo (https://weblogo.berkeley.edu/). Abundance plots in **g** created with BioRender.com.

Previous studies have identified V28 as a key residue but failed to characterize its role in bystander reduction[6,25,26]. We integrated thousands of pathogenic point mutations that are correctable by adenine base editing in HEK293T cells (Supplementary Fig. 5a,b). Overall, V28C demonstrated an increase of ~20% in on-target activity (A_6) and a 2.5- to 3-fold increase in precision when the editing context contained more than two or three adenines, respectively (Fig. 4a–c and Supplementary Fig. 5b–f).

Other variants that demonstrated high performance at the *DNMT1* site, including L34W, D54T and I95D, also exhibited enhanced precision but with reduced on-target efficiency (Fig. 4a–c). Similarly, M151E improved on-target efficiency but with a wider editing window than to V28C (Fig. 4a–c and Supplementary Fig. 5b–f). The addition of the M151E mutation to the V28C variant significantly impacted the on-target efficiency, achieving editing patterns and efficiencies similar to those of ABE9 (Supplementary Fig. 5b–f). Although these variants improved precision (for example, as reported in ABE9-related studies[10,13]), the associated reduction in editing efficiency limits their applicability across diverse targets and poses a significant challenge for in vivo therapeutic use.

To validate the library-based findings, we benchmarked V28C against ABE8e across 12 different endogenous sites in the human genome using HEK293T cells (Fig. 4d and Supplementary Fig. 5g). V28C refined the deaminase's editing window, improving precision at every tested site (Fig. 4d). Notably, it also exhibited higher editing activity at the most frequently edited adenine (Fig. 4d). V28C produced a constrained 4-A editing window, yielding an average 27.1% increase in on-target efficiency (Fig. 4e,f). The V28C mutation has been associated previously with elevated levels of C-to-T deamination[25]. To directly assess this in our system, we analyzed C-to-N editing across all tested loci using high-throughput sequencing. We did not observe a significant increase in C-to-T deamination relative to ABE8e (Supplementary Fig. 5h,i and Supplementary Table 3). These findings suggest that, in our experimental context, the V28C substitution does not substantially elevate cytosine bystander

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activity. Furthermore, our enzyme did not increase indel byproducts (Supplementary Fig. 5j).

ABE8e has been reported to favor deamination in YA contexts (Y = pyrimidine; T or C), whereas RA (R = purine; A or G) contexts are more challenging substrates[16]. Recently developed variants have improved potency and expanded sequence compatibility[16]. V28C showed improved editing efficiency compared to ABE8e-SpRY in RA contexts across our library testing (Supplementary Fig. 6a) and endogenous sites 12 and 16 (Fig. 4d and Supplementary Figs. 5g and 6b–d).

To determine whether this precision-enhancing effect is generalizable to other TadA variants, we introduced the V28C mutation into both ABE8e-V106W[19] (a variant previously reported to exhibit lower off-target activity) and CBE6b[27] (a CBE derived from TadA deaminase). Although the addition of V28C preserved the bystander reduction effect in both cases, it resulted in reduced editing efficiency in CBE6b_V28C (Supplementary Fig. 6e–g). These findings suggest that the structural tuning conferred by V28C can enhance BE precision across different deaminase scaffolds, but with trade-offs in activity.

Our evolved variants significantly reduced bystander editing while preserving robust editing activity. V28C consistently outperformed ABE8e by enhancing precision without widening the editing window. These results highlight the strength of our selection strategy in fine-tuning BE specificity, striking an optimal balance between efficiency and accuracy, critical for advancing therapeutic base editing applications.

V28C improves precise correction of pathogenic variants in induced pluripotent stem cells

NGS abundance analysis revealed that the V28C mutation significantly enriched single-base deamination across several loci. Structural modeling of the wild-type enzyme predicts that H57 and C87 establish three van der Waals interactions and two hydrogen bonds with 8-Az(26) (Fig. 5a). In the V28C mutant, we hypothesize that C28 shifts closer to 8-Az(26), reducing the measured distance to below 5.101 Å. In line with this, we detected a 0.77 Å decrease in distance between residue 28 and 8-Az(26) (Fig. 5a). This structural rearrangement probably constrains the active site, enhancing substrate specificity while reducing off-target deamination. By restricting the flexibility of the deaminase domain, V28C narrows the editing window and minimizes bystander editing without compromising catalytic efficiency.

To assess the functional impact of this enhanced precision, we tested the V28C variant in generating a splicing disruption in *PCSK9*—a therapeutic target for lowering LDL levels and reducing the risk of coronary heart disease[28]. Using a previously validated gRNA (NGG PAM) targeting the exon 1–intron 1 splice donor site[29], we introduced a loss-of-function mutation (Fig. 5b). Here, even in the absence of bystander editing, V28C-WT exhibited an average editing efficiency of 75.7% (±5.0), outperforming ABE8e-nSpCas9-WT by 19.7% (Fig. 5c,d).

To further evaluate the precision of V28C in a clinically relevant setting, we targeted the SNCA E46K mutation, which causes early-onset Parkinson's disease by promoting α-synuclein aggregation and neuronal toxicity (Fig. 5e). The target sequence shares 45% identity with the *DNMT1* site used to identify agRNA$_{56114}$ (Fig. 5f). As expected, V28C demonstrated superior precision in reverting the pathogenic mutation, achieving 11.6% (±0.8) perfect edits (percentage of edited reads) compared to just 0.65% (±0.14) with ABE8e, representing a 21.1-fold increase in precision (±1.7) (of total reads) (Fig. 5j). The combination of V28C with agRNA$_{56114}$ further improved precision, yielding 17.5% (±1.06) perfect edits, a 37.6-fold enhancement (±2.5) over ABE8e alone (Fig. 5i,j). Although introducing the PLM predicted M151E mutation into V28C further increased perfect editing to 31.1% (±1.4) when combined with sgRNA$_{Ctrl}$ and 34.8% (±0.7) with agRNA$_{56114}$ (calculated from edited reads), this came at the cost of significantly reduced overall efficiency (Supplementary Fig. 7a–c). V28C showed a significant reduction on indel formation and no significant changes in Cas9-dependent off targets (Supplementary Fig. 7d,e).

These findings underline the power of PANCE to design BEs with enhanced precision and minimize bystander editing. Moreover, the synergy between V28C and agRNA strategies suggests a tunable approach to optimizing editing outcomes. Further investigation into understanding the basis of anchor-driven bystander reduction will be critical to fully harnessing these tools for therapeutic applications.

Discussion

Our findings highlight an advancement in base editing precision, achieved through the V28C variant, which enhances target specificity while preserving efficiency. By minimally altering the deaminase active site, the V28C mutation restricts the editing window, effectively reducing bystander mutations without compromising catalytic activity. Whereas V28C falls within a previously described mutational hotspot[6,26,27,30], several other substitutions uncovered by our PANCE circuit (including L34W, D54T and I95D) were not reported previously. These findings highlight the power of an unbiased evolutionary system to identify functionally relevant variants that conventional approaches, such as saturation mutagenesis or rational design[6–8], might miss. Notably, several of these new variants reside in structural domains of TadA-8e[18], suggesting that changes outside known hotspots can meaningfully influence base editing precision. Together, these results demonstrate that PANCE enables systematic exploration of both canonical and noncanonical sites, providing a more comprehensive route to optimize precise BEs. The diverse set of variants identified here can serve as a valuable dataset for training future ML models, further accelerating the rational design of alternative editors. Moreover, future modifications of our selection circuit or the application of continuous evolution (PACE)[20] may enable the discovery of additional variants with enhanced precision and efficiency.

By extending the 3′-end of gRNAs we discovered a new gRNA modification that can influence the editing window and efficiency. Although agRNA$_{56114}$ proved to be beneficial in contexts with sequence similarity to *DNMT1*, most new targets will require a new custom agRNA (Supplementary Fig. 8a,b). To find universal rules for the design of agRNAs for new contexts, more data and, potentially, a crystal structure of the agRNA-target complex will be necessary.

Using PLM, we detected a potential functional redundancy of certain ABE8e mutations. Furthermore, we detected new mutations (M151E) that increase precision and efficiency of ABE8e.

The therapeutic relevance of these improvements is evident in the correction of highly challenging pathogenic mutations, such as the SNCA E46K variant. V28C achieved a 21.1-fold increase in precise editing compared to ABE8e—a result further enhanced by pairing with agRNA$_{56114}$, which boosted precision 37.6-fold. This strategy offers an alternative path forward for correcting mutations in clinically relevant settings, where even minor bystander edits could have functional consequences. The ability to enhance on-target accuracy without sacrificing efficiency marks an important step toward developing safer genome-editing therapies for genetic disorders. The panel of BEs derived from this study can be integrated easily into current state-of-the-art therapeutic strategies[31]. Although the agRNA includes a 3′ extension that increases its length relative to a standard gRNA, its size is comparable to pegRNAs used in prime editing, which have been synthesized and delivered successfully using lipid nanoparticles in both preclinical and therapeutic contexts[32,33].

The principles used here, leveraging structural insights and directed evolution, can be applied to additional enzyme classes, including CBEs, to broaden the scope of precise genetic corrections. Future work will focus on understanding the molecular basis of anchor-driven precision and testing these evolved editors in disease models. By fine-tuning the balance between efficiency and specificity, this approach pushes genome editing closer to clinical application, offering a powerful tool for precision medicine.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-025-02937-w.

References

1. Rees, H. A. & Liu, D. R. Base editing: precision chemistry on the genome and transcriptome of living cells. *Nat. Rev. Genet.* **19**, 770–788 (2018).
2. Gaudelli, N. M. et al. Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage. *Nature* **551**, 464–471 (2017).
3. Gaudelli, N. M. et al. Directed evolution of adenine base editors with increased activity and therapeutic application. *Nat. Biotechnol.* **38**, 892–900 (2020).
4. Walton, R. T., Christie, K. A., Whittaker, M. N. & Kleinstiver, B. P. Unconstrained genome targeting with near-PAMless engineered CRISPR–Cas9 variants. *Science* **368**, 290–296 (2020).
5. Wang, J. Y. & Doudna, J. A. CRISPR technology: a decade of genome editing is only the beginning. *Science* **379**, eadd8643 (2023).
6. Chen, L. et al. Engineering a precise adenine base editor with minimal bystander editing. *Nat. Chem. Biol.* **19**, 101–110 (2023).
7. Jeong, Y. K. et al. Adenine base editor engineering reduces editing of bystander cytosines. *Nat. Biotechnol.* **39**, 1426–1433 (2021).
8. Kim, Y. B. et al. Increasing the genome-targeting scope and precision of base editing with engineered Cas9-cytidine deaminase fusions. *Nat. Biotechnol.* **35**, 371–376 (2017).
9. Zhao, N. et al. Evolved cytidine and adenine base editors with high precision and minimized off-target activity by a continuous directed evolution system in mammalian cells. *Nat. Commun.* **15**, 8140 (2024).
10. Hsiao, S. et al. Library-assisted evolution in eukaryotic cells yield adenine base editors with enhanced editing specificity. *Adv. Sci. (Weinh.)* **11**, e2309004 (2024).
11. Zhao, D. et al. Imperfect guide-RNA (igRNA) enables CRISPR single-base editing with ABE and CBE. *Nucleic Acids Res.* **50**, 4161–4170 (2022).
12. Alves, C. R. R. et al. Optimization of base editors for the functional correction of SMN2 as a treatment for spinal muscular atrophy. *Nat. Biomed. Eng.* **8**, 118–131 (2023).
13. Qin, W. et al. ABE-ultramax for high-efficiency biallelic adenine base editing in zebrafish. *Nat. Commun.* **15**, 5613 (2024).
14. Arbab, M. et al. Determinants of base editing outcomes from target library analysis and machine learning. *Cell* **182**, 463–480 (2020).
15. Zhao, L. et al. PAM-flexible genome editing with an engineered chimeric Cas9. *Nat. Commun.* **14**, 6175 (2023).
16. Xiao, Y.-L., Wu, Y. & Tang, W. An adenine base editor variant expands context compatibility. *Nat. Biotechnol.* **42**, 1442–1453 (2024).
17. Miller, S. M., Wang, T. & Liu, D. R. Phage-assisted continuous and non-continuous evolution. *Nat. Protoc.* **15**, 4101–4127 (2020).
18. Lapinaite, A. et al. DNA capture by a CRISPR–Cas9-guided adenine base editor. *Science* **369**, 566–571 (2020).
19. Richter, M. F. et al. Phage-assisted evolution of an adenine base editor with improved Cas domain compatibility and activity. *Nat. Biotechnol.* **38**, 883–891 (2020).
20. Esvelt, K. M., Carlson, J. C. & Liu, D. R. A system for the continuous directed evolution of biomolecules. *Nature* **472**, 499–503 (2011).
21. Weiss, G. A., Roth, T. A., Baldi, P. F. & Sidhu, S. S. Comprehensive mutagenesis of the C-terminal domain of the M13 gene-3 minor coat protein: the requirements for assembly into the bacteriophage particle. *J. Mol. Biol.* **332**, 777–782 (2003).
22. Doman, J. L., Raguram, A., Newby, G. A. & Liu, D. R. Evaluation and minimization of Cas9-independent off-target DNA editing by cytosine base editors. *Nat. Biotechnol.* **38**, 620–628 (2020).
23. Hie, B. L., Yang, K. K. & Kim, P. S. Evolutionary velocity with protein language models predicts evolutionary dynamics of diverse proteins. *Cell Syst.* **13**, 274–285 (2022).
24. Rives, A. et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proc. Natl Acad. Sci. USA* **118**, e2016239118 (2021).
25. Chen, L. et al. Re-engineering the adenine deaminase TadA-8e for efficient and specific CRISPR-based cytosine base editing. *Nat. Biotechnol.* **41**, 663–672 (2023).
26. Cho, S.-I. et al. Engineering TALE-linked deaminases to facilitate precision adenine base editing in mitochondrial DNA. *Cell* **187**, 95–109 (2024).
27. Zhang, E., Neugebauer, M. E., Krasnow, N. A. & Liu, D. R. Phage-assisted evolution of highly active cytosine base editors with enhanced selectivity and minimal sequence context preference. *Nat. Commun.* **15**, 1697 (2024).
28. Lepor, N. E. & Kereiakes, D. J. The PCSK9 inhibitors: a novel therapeutic target enters clinical practice. *Am. Health Drug Benefits* **8**, 483–489 (2015).
29. Musunuru, K. et al. In vivo CRISPR base editing of PCSK9 durably lowers cholesterol in primates. *Nature* **593**, 429–434 (2021).
30. Neugebauer, M. E. et al. Evolution of an adenine base editor into a small, efficient cytosine base editor with low off-target activity. *Nat. Biotechnol.* **41**, 673–685 (2023).
31. Musunuru, K. et al. Patient-specific in vivo gene editing to treat a rare genetic disease. *N. Engl. J. Med.* **392**, 2235–2243 (2025).
32. Anzalone, A. V. et al. Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature* **576**, 149–157 (2019).
33. Bae, S., Park, J. & Kim, J.-S. Cas-OFFinder: a fast and versatile algorithm that searches for potential off-target sites of Cas9 RNA-guided endonucleases. *Bioinformatics* **30**, 1473–1475 (2014).

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Methods

Cell lines and cell culture

HEK293T (CRL-3216), HepG2 (HB-8065) and HeLa (CCL-2) cell lines were obtained from the American Type Culture Collection. Cells were maintained in Dulbecco's Modified Eagle Medium (DMEM; Gibco, cat. no. 11995073) supplemented with 10% heat-inactivated fetal bovine serum (FBS; Gibco, cat. no. A56708-01) and 1% penicillin-streptomycin (Gibco, cat. no. 15070063) at 37 °C in a humidified atmosphere containing 5% CO_2.

The KOLF2.1J SNCA E46K$^{-/-}$ induced pluripotent stem (iPS) cell line was purchased from the Jackson Laboratory and cultured in StemFlex medium (Life Technologies, cat. no. A3349401) supplemented with 10% FBS (Life Technologies). iPS cells were plated on dishes coated with Synthemax II-SC (1 mg ml^{-1} stock solution; Corning, cat. no. 3535) according to the manufacturer's instructions and treated with 10 µM Y-27632 for 24 h following each passage.

All cultures were tested routinely for Mycoplasma contamination every 3 months using the American Type Culture Collection Universal Mycoplasma Detection Kit, and all tests were negative. For passaging, adherent cell lines were dissociated using TrypLE Express (Gibco, cat. no. 12604013) for 3–5 min, whereas iPS cells were detached gently with ReLeSR (Stem Cell Technologies, cat. no. 100-0483). For experiments requiring single-cell suspensions, iPS cells were detached using Accutase (Stem Cell Technologies, cat. no. 07922) and counted using trypan blue exclusion on a Countess II automated cell counter (Thermo Fisher Scientific).

Bacterial media, reagents and plasmids. Luria broth (LB) and 2× YT medium were prepared using MP Biomedicals media capsules according to the manufacturer's protocol. For LB and 2× YT agar, 16 g l^{-1} agar was added for standard, and 7 g l^{-1} agar was added for soft agar. All media was sterilized by autoclaving. Oligonucleotides, primers and plasmids used in the study can be found in Supplementary Table 4. All gRNAs were cloned using KLD (NEB, cat. no. M0554S) according to the manufacturer's protocol and spacer sequences are in Supplementary Table 9.

agRNA library generation. The agRNA library consists of an upstream binding sequence that is the reverse complement of the downstream sequence of the target, a counter-loop and a downstream binding sequence that binds the upstream sequence of the target. The upstream and downstream binding sequences are of different length and have different binding regions in the 1–11 bp region upstream and downstream of the target. The counter-loop library includes 33 different DNA sequences of which the longer sequences form GC-rich hairpins. The length of the counter-loop ranges from 1 to 14 nucleotides. The final library contains every combination of the possible upstream binding sequence, counter-loop and downstream binding sequence combinations. A script to generate the hairpin library for a new context can be found in Supplementary Code 1. To facilitate agRNA library design for new targets, we developed a Python script (Supplementary Code 1) that automatically generates library sequences tailored to any custom base editing site.

The agRNA for *DNMT1* was ordered as an Agilent DNA Oligo Pool (64,610 oligonucleotides; Supplementary Tables 5 and 6). The oligonucleotides for the *DNMT1* library contained a Gibson overhang, gRNA, gRNA scaffold, agRNA library and a terminator followed by a short DNA sequence used as primer binding site. The target for the *DNMT1* library was already cloned on the plasmid used as a backbone. The DNA Oligo Pool library was amplified by PCR with the oligonucleotides Lib_F and Lib_R using Q5 Hot Start High-Fidelity DNA Polymerase (NEB, cat. no. M0493S). The backbone pU6-tevopreq1-GG-acceptor (Addgene, cat. no. 174038) was PCR amplified using the oligos SplitF and SplitR. The PCR product of the backbone was digested overnight with *Dpn*I (NEB, cat. no. R0176S) at 37 °C and both PCR products were purified using the Monarch PCR and DNA Cleanup Kit (NEB, cat. no. T1130S) according to the manufacturer's protocol. The fragments were assembled using the Gibson Assembly Master Mix (NEB, cat. no. E2611S) in a 10:1 ratio library:backbone and 150 ng of the backbone DNA according to the manufacturer's protocol. A 2-µl aliquot of the Gibson assembly mix was used to directly transform DUOs electrocompetent cells (Endura, cat. no. 60242-1) and, after recovery on 1 ml of SOC medium, plated on Carbenicillin/Agar plates poured in Nunc Square BioAssay Dishes (Cole Palmer, cat. no. EW-01929-00). For cloning into the backbone LentiGuide-Puro (Addgene, cat. no. 52963), the library was amplified from the pU6-tevopreq1-GG-acceptor. The backbone was digested using *Psp*XI (NEB, cat. no. R0656S) and Esp3I (NEB, cat. no. R0734S) and cloned by Gibson assembly following the previously described protocol.

For each library at least ten times library coverage of colonies were washed off the plates using LB medium then spun down. The plasmid DNA was extracted from the resulting pellet using a Plasmid Plus Midi Kit (Qiagen, cat. no. 12943) according to the manufacturer's protocol.

Lentiviral particles generation. HEK293T were cells were seeded at a density of 5×10^6 cells per 10-cm plate in DMEM supplemented with 10% FBS and antibiotics (Thermo Fisher, cat. no. 15240062). The following day, cells were transfected using Transporter 5 Transfection Reagent (Polysciences, cat. no. 26008) with a plasmid mix containing pVSV-G (3.86 µg), pPax2 (8.57 µg) and the lentiviral transfer vector (9.23 µg) in Opti-MEM (Thermo Fisher). The DNA-transporter complexes were incubated at room temperature for 20 min before being added to the culture medium. After 24 h, the medium was replaced with fresh DMEM, and viral supernatants were collected at 48- and 72-h post-transfection. The harvested medium was filtered through a 0.45-µm vacuum filter system and concentrated using Lenti-X Concentrator (Takara Bio, cat. no. 631231) at a 1:3 ratio (media to concentrator) by incubation at 4 °C for at least 30 min, followed by centrifugation at 1,500*g* for 45 min at 4 °C. The viral pellet was resuspended in phosphate-buffered saline, aliquoted and stored at −80 °C until further use.

***DNMT1* library testing in HEK cells.** HEK293T cells were transduced with the lentiviral library with a multiplicity of infection of 0.2. After 24–48 h, medium was removed and exchanged for fresh medium with 2 µg ml^{-1} of Puromycin. Selection continued for 2 weeks. To test the editing pattern across our library, 20 million cells (300× coverage) were seeded in a 225 mm^3 dish and transfected the day after using Lipofectamine 3000 (Invitrogen, cat. no. L3000015) with 20 µg of pCMV-T7-ABE8e-nSpCas9-P2A-EGFP (KAC978) (Addgene, cat. no. 185910). Genomic DNA was collected from cells 5 days after transfection.

Evaluation of the anchor library. Efficiency and precision of the BE in combination with the agRNA were evaluated using custom R script. The quality of the reads from NGS samples was assessed before further processing. Variant calling techniques were then applied to distinctly identify the perfect edit (conversion of adenine at position 8 to guanine) apart from bystander edits, which encompassed any conversion of the other adenines or combinations involving A_8. Samples with anchor sequences yielding fewer than 20 reads were excluded to ensure robustness in the data analysis. Furthermore, a quantitative score was devised and calculated using the following formula:

$$\text{Score} = (\,\%\,\text{PerfectEdit})/((\,\%\,\text{PerfectEdit} + \,\%\,\text{Bystander})^2)$$

Anchors achieving the highest scores and demonstrating at least 20% overall editing efficiency were further characterized experimentally (Supplementary Tables 7 and 8).

Context library generation. For the A-to-G BE, ~12,000 different gRNAs targeting pathogenic relevant mutations[14] were cloned as a library to test the performance of different hairpins and BEs.

The generation of these context libraries differed from the generation of the agRNA, as extensive recombination events occurred when the gRNA, gRNA scaffold, agRNA and target were introduced as one oligonucleotide. To avoid recombination issues, the gRNA and target with 11 bp upstream and 25 bp downstream of the native genomic context were cloned as an oligo lacking the gRNA scaffold and hairpin. Instead of these, the oligo had two outwards-facing *Bsa*I-HF (NEB, cat. no. R3733S) cutting sites with ten randomized base pairs at that position. The DNA Oligo Pool libraries are amplified using PCR with the oligonucleotides Lib_F and Lib_R. The backbone sgBbsI (p2Tol-U6-2×BbsI-sgRNA-HygR) (Addgene, cat. no. 71485) was PCR amplified using the oligos BB_R and BB_F. The PCR product of the backbone was digested with *Dpn*I overnight at 37 °C and both PCR products were purified using the NEB Monarch PCR and DNA Cleanup Kit according to the manufacturer's protocol. The fragments were assembled using the New England Biolabs Gibson Assembly Master Mix in a 10:1 library:backbone ratio and 150 ng of the backbone DNA according to the manufacturer's protocol. A 2-µl aliquot of the Gibson assembly mix was used directly to transform Lucigen's Endura Competent Cells and, after recovery, plated on agar plates poured in Nunc Square BioAssay Dishes. For each library at least ten times library coverage of colonies were washed off the plates using Luria broth medium and then spun down. The plasmid DNA was extracted using the QIAGEN Plasmid Plus Midi Kit according to the manufacturer's protocol from the resulting pellet.

The library was then digested using BsaI according to the manufacturer's protocol and gel purified using the New England Biolabs Monarch Gel Extraction Kit according to the Manufacturer's protocol. The gRNA scaffold, hairpin and terminator with an inward facing BsaI cutting site up- and downstream were ordered as cloned gene synthesis from IDT. The plasmid was also BsaI digested, and the fragment was purified using the New England Biolabs Monarch Gel Extraction Kit according to the Manufacturer's protocol. The insert and library were ligated using New England Biolabs T4 DNA Ligase according to the Manufacturer's protocol with 150 ng backbone and a 10:1 ratio of the insert to the backbone. 2 µl of the ligation mix were directly transformed into Lucigen's Endura Competent Cells and after recovery plated on agar plates poured in Nunc Square BioAssay Dishes. For each library at least 10x library coverage of colonies were washed off the plates using LB medium and then spun down. The plasmid DNA was extracted from the resulting pellet using the QIAGEN Plasmid Plus Midi Kit according to the manufacturer's protocol.

Path_Var library testing in HEK cells. For stable Tol2 transposon-mediated library integration, 5 million HEK293T cells (~400× coverage) were seeded in 175 mm^3 dishes. The following day, cells were cotransfected using Lipofectamine 3000 with 10 µg of Tol2 transposase plasmid (pCMV-Tol2; Addgene, cat. no. 31823) and 10 µg of Path_Var library. To generate stable library cell lines, cells were selected with hygromycin (25 µg ml^{-1}) starting the day after transfection and continued for 2 weeks. Thereafter, 2.5 million cells were transfected with 10 µg of BE using Lipofectamine 3000 to (~200× coverage) were seeded the day before in a 100-mm^3 dish. Genomic DNA was collected from cells 5 days after transfection.

Illumina sequencing and bioinformatic analysis of the libraries. To sequence the libraries before (as quality control) and after testing in the HEK293T cells, 1 ng of the isolated plasmid DNA (QuickExtract DNA Extraction Solution Biosearch Technologies) was amplified using the oligonucleotide mix IllSeq_DNMT1_i5_F1-4 and IllSeq_DNMT1_i7_R1-4 using Q5 High-Fidelity 2× Master Mix according to the manufacturer's protocol. The forward and reverse oligonucleotide contained an i5/i7 overhang for indexing as well as four to seven Ns to ensure shifting of the sequence to be able to determine sequences with high identity. The resulting PCR products were amplified in a second PCR reaction using a compatible combination of the NEBNext Multiplex Oligos for Illumina (NEB, cat. no. E6440S). The PCR products were purified using the Monarch Gel Extraction Kit (NEB, cat. no. T1120S) and quantified using Qubit-HS (Invitrogen, cat. no. Q33231). A 4-nM pool of the different libraries was generated and 10% 4 nM PhiX was added. The pool was sequenced using the Illumina MiSeq Reagent Kit v.2 (300 cycles) according to the manufacturer's protocol.

Evaluation of the context library. Evaluation of the context library involved analyzing gRNA libraries, which comprised approximately 12,000 spacer sequences and their respective contextual sequences. The efficiency and editing profiles for each gRNA were established using custom scripts developed in R. First, the target sites, where each spacer binds within the context, were extracted from the NGS reads. Subsequently, for each spacer in the library, all combinations of adenine to guanine conversions were aligned against these extracted sequences. Spacers with fewer than 25 total reads were excluded from the analysis. To quantify overall editing efficiency for the different BEs, the mean A to G conversion rate was calculated by averaging the editing frequencies at each targeted position.

Genome editing of endogenous human genomic loci. A total of 30,000 cells (HEK293T, HeLa and HepG2) were seeded in 96-well plates (Corning) and transfected the next day using jetOPTIMUS (Sartorius, cat. no. 101000006) or TransIT-X2 (MirusBio, 6003) following the manufacturer instructions. Then, 50 ng of sgRNA or agRNA (both cloned in BPK1520; Addgene, cat. no. 65777) with 150 ng of BE variant were cotransfected, and cells were harvested after 3 days for Sanger sequencing (Azenta Life Sciences) or NGS (Quintara Biosciences or in-house Illumina miSeq). Cas9-dependent genomic off-target sites, predicted using Cas-OFFinder[32], were analyzed by targeted PCR by NGS. Spacer sequences of tested loci are found in Supplementary Table 9. NGS data were analyzed using CRISPRESSO[33] and BE-analyzer (CRISPR RGEN tools)[33] and representative substitution tables are found in Supplementary Table 3. gRNAs used in Fig. 4d and Supplementary Figs. 5g, 6b,e,f and 8b were described previously in ref. 34. gRNAs used in Supplementary Fig. 6g were described previously in ref. 16. gRNAs used in Supplementary Fig. 6b were described previously in ref. 35.

DNA Cas9-independent off-target editing analysis

The analysis of Cas9-independent off-target analysis was performed by R-loop assay following previously detailed protocols and gRNAs[22]; 150 ng of BE (nSpCas9-ABE8e-SpRY or evolved variant), 100 ng of SpCas9 gRNA plasmid, 150 ng of a catalytically dead SaCas9 and 100 ng of SaCas9 gRNA plasmid (targeting a genomic locus unrelated to the on-target site) were cotransfected into HEK293T cells. DNA was extracted after 72 h and analyzed using high-throughput sequencing.

RNA sequencing

HEK293T cells were seeded in 48-well plates and transfected with 300 ng of BE plasmid and 100 ng of gRNA plasmid per well. After 48 h, cells were lysed and total RNA was extracted using TRIzol (Invitrogen, cat. no. 15596026) reagent according to the manufacturer's instructions. RNA sequencing was performed by Azenta Life Sciences according to the following protocol.

For RNA sequencing with poly(A) selection, total RNA was quantified using a Qubit 2.0 Fluorometer (Life Technologies), and RNA integrity was assessed using the Agilent TapeStation 4200 (Agilent Technologies). ERCC RNA Spike-In Mix (Thermo Fisher Scientific) was added to normalized RNA samples following the manufacturer's protocol.

RNA sequencing libraries were prepared using the NEBNext Ultra II RNA Library Prep Kit for Illumina (NEB, cat. no. E7770S), following the manufacturer's instructions. Briefly, mRNA was enriched using oligo(dT) beads, then fragmented at 94 °C for 15 min. First- and second-strand cDNA synthesis was performed, followed by end repair, 3′ adenylation, adapter ligation, index addition and PCR enrichment. Libraries were validated using the Agilent TapeStation and quantified using the Qubit 2.0 Fluorometer and quantitative PCR (KAPA Biosystems).

Prepared libraries were multiplexed and sequenced on an Illumina NovaSeq X Plus system (Illumina) using 2 × 150 bp paired-end reads. Image analysis and base calling were performed using NovaSeq Control Software (NCS), and raw BCL files were converted to FASTQ format and demultiplexed using bcl2fastq v.2.20, allowing one mismatch in index sequences.

RNA Cas9-independent off-target editing analysis

To assess A-to-I RNA editing, SNP and indel calling was performed using Samtools mpileup (v.1.3.1) followed by VarScan (v.2.3.9) with the following parameters: –min-coverage 1–min-reads2 1–min-var-freq 0.01–*P* value 0.05. Variant call format files from BE-treated samples were analyzed for A-to-I RNA editing events using a custom pipeline. Control variants were collected from three independent control samples and merged to generate a comprehensive background set (40,625 variants), which was subtracted from treatment samples to exclude germline or systematic variants.

A-to-G substitutions were considered putative A-to-I editing sites. Stringent quality filters were applied: minimum read depth ≥5×, minimum allele frequency ≥1% and maximum allele frequency ≤99% (to remove homozygous SNPs). After background subtraction, high-confidence A-to-I editing sites were retained. Editing percentages were calculated per site as the ratio of edited reads to total reads (AD/DP in the variant call format) and per sample as the average editing frequency across all sites passing filters.

Using this approach, we identified 207,541 high-confidence A-to-I editing sites across 12 samples (ABE8e-SpRY, V28C-SpRY and L34W-SpRY BEs with $sgRNA_{Ctrl}$ or $agRNA_{56114}$, each in duplicate).

Generation of the selection phages for PANCE. The PANCE selection phages are carrying the CDS for the ABE8e adenine deaminase instead of the CDS of PIII. The ABE8e adenine deaminase has part of the peptide linker sequence and a C-terminal fused intein CDS to enable it to encode the relatively small protein and not the whole BE. The phages were generated by PCR amplifying the ABE8e adenine deaminase including the partial sequence of the peptide linker using the oligonucleotides ABE_M13_F and ABE_M13_R. The N-terminal Npu DnaE intein was ordered as gBlock and amplified using the oligonucleotides Npu_ABE_F and Npu_M13_R. The phage backbone was amplified using the oligonucleotides GOI_M13_F and GOI_M13_R using M13KO7 helper phage (NEB, cat. no. N0315S) genomic DNA as a template. All PCRs were performed using Q5 High-Fidelity 2× Master Mix according to the manufacturer's protocol using 1 ng of template DNA and an annealing temperature of 60 °C. All fragments were digested with *Dpn*I (NEB) overnight at 37 °C in the PCR buffer and PCR purified using the NEB Monarch PCR and DNA Cleanup Kit the next day. The fragments were assembled using the NEB Gibson Assembly Master Mix according to the manufacturer's protocol and 3 µl of the reaction were transformed directly into electrocompetent S2060 pJC175e competent cells. The cells were recovered in 500 µl SOC medium for 45 min, then 450 µl and 50 µl were each mixed with 500 µl freshly grown S2060 pJC175e cells. The cells were mixed immediately with 3 ml soft LB agar (0.7%) and plated on LB bottom agar plates containing 100 µg ml^{-1} carbenicillin. The plates were incubated at 37 °C overnight. Plaques were picked into 50 µl 2× YT medium and 1 µl was used as a template for colony PCR using the oligonucleotides ABE_M13_F and Npu_M13_R. Positive phages were amplified by adding the remaining 2× YT medium to a freshly grown S2060 pJC175e culture at the OD_{600} of 0.4 and cultivating for 16 h at 37 °C. The cultures were spun down to remove the *Escherichia coli* cells and the phages were precipitated by adding a 20% polyethylene glycol (8000) and 2.5 M sodium chloride solution in a 1:4 ratio to the culture supernatant. The mixture is incubated for at least 3 h at 4 °C and the phage pellet is resuspended in a phosphate-buffered saline buffer, the phage titer was quantified using the Phage Titration ELISA kit (Progen, PRPHAGE) and the phages were stored at 4 °C until use. Furthermore, 3 ml of the culture supernatant was used for phage DNA isolation using the E.Z.N.A. M13 DNA Mini kit (Omega Bio-tek, cat. no. D6900-01). The isolated DNA was sent to Plasmidsaurus for whole-phage DNA sequencing.

Generation of the selection cells for PANCE. The selection plasmids were designed on the basis of using pJC175e (Addgene, cat. no. 79219)[36] and adding mutations that, when edited by the ABE, only perfect edits restore PIII activity whereas bystander edits lower pIII activity. The pJC175e backbone was amplified using the oligonucleotides pIII_gBlock_R and pJC175e_Cas_F. The part of the pIII CDS containing the mutation followed by the corresponding guide correcting the introduced mutation downstream as well as the C-terminal DnaE intein necessary to fuse the ABE8e encoded by the phage to the nSpCas9-SpRY encoded by the selection plasmid were ordered as gBlock. Each selection plasmid also encodes the agRNA to fix the mutation on pIII (Supplementary Table 8). The three different gBlocks for the three different selection plasmids each encoding a different pIII mutation were amplified by PCR using the oligonucleotides gBlock_R and gBlock_pIII_F. The base nSpCas9 CDS was amplified from ABE8e plasmid (Addgene, cat. no. 138489) using the oligonucleotides BE_Npu_F and BE_pJC175e_R. All PCRs were performed using Q5 High-Fidelity 2× Master Mix according to the manufacturer's protocol using 1 ng of template DNA and an annealing temperature of 60 °C. All fragments were digested with *Dpn*I overnight at 37 °C in the PCR buffer and PCR purified using the NEB Monarch PCR and DNA Cleanup kit the next day. The fragments were assembled using the NEB Gibson Assembly Master Mix according to the manufacturer's protocol and transformed into electrocompetent S2060 competent cells. The cells were recovered in 500 µl SOC medium for 1 h then plated on LB agar plates with 100 µg ml^{-1} carbenicillin and incubated overnight at 37 °C. Colonies were screened using colony PCR and positive clones were sent for whole-plasmid sequencing. Clones with verified sequence were used to generate electrocompetent cells that were then transformed with the mutation plasmid MP4 (Addgene, cat. no. 69652)[37]. The cells were recovered in 1 ml SOC medium and plated on 2× YT agar plates containing 1% glucose, 100 µg ml^{-1} carbenicillin and 25 µg ml^{-1} chloramphenicol. Five colonies were used to start 50-µl shake flask 2× YT 1% glucose, 100 µg ml^{-1} carbenicillin and 25 µg ml^{-1} chloramphenicol cultivations. The cultivations were used to freeze 20% glycerol stocks in 1 ml aliquots after 16 h. Each culture was also used to isolate plasmid DNA for whole-plasmid sequencing by Plasmidsaurus to select the glycerol stocks with no mutation in MP4 and the selection plasmid.

PANCE. The evolution was performed as ten consecutive batch cultivations in triplicate using a mix of three different selection plasmids in each evolution. For the PANCE experiment, the day before the cultivation, three 3-ml overnight cultures were prepared using 2× YT medium with 1% glucose, 100 µg ml^{-1} carbenicillin and 25 µg ml^{-1} chloramphenicol. The cultures are inoculated with the glycerol stock of one of the selection plasmids each. The following day, 3–4 h before phage infection, 50-ml shake flasks are inoculated with a combined OD_{600} of 0.1 of the pooled overnight cultures with the different selection plasmids. The cells are cultivated in 2× YT with 100 µg ml^{-1} carbenicillin and 25 µg ml^{-1} chloramphenicol. At 30 min before reaching an OD_{600} of 0.4, the cells are induced with 0.5% arabinose and when the cells reach the OD_{600} of 0.4, the cells are infected with the selection phages at a multiplicity of infection of 1 for the first selection round. The evolution is performed for 12 h at 37 °C; thereafter, the entire cultivation was spun down and the supernatant was filtered with 0.2 µM filters. For selection rounds

2–4, 500 µl, for rounds 5–6 100 µl and for the remaining rounds 5 µl of the supernatant were used to infect the following evolution. The phage titer after each selection round was determined using the Progen Phage Titration ELISA kit. A 3-ml aliquot of each culture supernatant was used for phage DNA isolation using the Omega Bio-tek E.Z.N.A. M13 DNA Mini Kit. Sequences of V28C, L34W, M151E and V28C-M151E are detailed in Supplementary Table 8.

Illumina sequencing of the PANCE experiment. To sequence the PANCE variants, 1 ng of the isolated plasmid phage DNA of each selection round was amplified using the oligonucleotide mix IllSeq_ABE_i5_F1-4 and IllSeq_ABE_i5_R1-4 using Q5 High-Fidelity 2× Master Mix according to the manufacturer's protocol. The forward and reverse oligonucleotide contained an i5/i7 overhang for indexing as well as four to seven Ns to ensure shifting of the sequence to be able to determine sequences with high identity. The resulting PCR products were amplified in a second PCR reaction using a compatible combination of the NEBNext Multiplex Oligos for Illumina. The PCR products were purified using the Monarch Gel Extraction Kit and quantified using Qubit-HS. A 4-nM pool of the different libraries was generated and 10% 4 nM PhiX was added. The pool was sequenced using the MiSeq Reagent Kit v.3 (600-cycles, Illumina, MS-102-3003) according to the manufacturer's protocol.

PreS calculation. To quantify the specificity and efficiency of each variant in editing the target nucleotide with minimal bystander activity, we defined a PreS using the following formula:

$$\text{PreS} = \left(\%\ \frac{\text{On target editing}}{\text{Average bystander editing}}\right) \times \left(\%\ \frac{\text{On target editing (variant)}}{\text{On target editing (ABE8eCtrl)}}\right)$$

This score integrates two key components:

Editing precision as the ratio of on-target editing to average bystander editing. This reflects the precision of editing at the intended nucleotide relative to unintended nearby edits.

Relative efficiency as the comparison of a variant's on-target editing efficiency to that of ABE8e—our reference editor. This normalization controls for differences in baseline editing efficiency and allows comparison across variants.

A higher PreS indicates a variant that edits the intended base with high efficiency and minimal bystander activity relative to ABE8e. For example, ABE8e-SpRY$_{\mathrm{Ctrl}}$ has a PreS of 0.8 at the *DNMT1* site. The final value was calculated by dividing the editing efficiency at position A8 (15.3%) by the total bystander editing at positions A4, A5 and A6 (19%), and multiplying by a normalization factor derived from the ratio of editing at A8 to itself (15.3%/15.3%), which simplifies to 1. All PreS above 0.8 indicate higher precision and or activity.

Evolutionary plausible mutations prediction with ML. We followed the approach used by Hie et al.[23], which consists of using an ensemble of six PLMs: ESM-1b[23] and ESM-1v[24], composed itself of five models (accessible at: https://github.com/facebookresearch/esm). Together, the six models are used to predict what amino acids, if any, would be more likely than the current wild-type amino acid at each position of the protein sequence given as input. The number of models in the ensemble agreeing on a given prediction (that is, a specific amino acid substitution) allows to score a given substitution, where a higher score is more likely to yield a positive result.

We applied the ensemble of PLMs to the TadA-8e sequence, which yielded the following predictions (PreS in parenthesis): R26G (6), F84L (6), N108D (6), Y149F (6), F156K (6), V106A (5), P152R (5), H8D (5), N157K (5), R111T (4), C146S (3), R111A (2), C146Q (2), M151E (2), C146K (1), Y123H (1), M151Q (1), A48P (1), V155E (1), S109P (1), P152Q (1).

Computational analysis of structural impact on ABE8e. We used the crystal structure of ABE8e named 6vpc available in the Protein Data Bank database and the software ChimeraX (v.1.6.1 (9 May 2023) for visualization and analysis of the structure.

We first attempted to predict the structures of relevant mutants in this study using AlphaFold2 (available in the colab notebook from sokrypton). However, no structural change was predicted for these single or double mutants with AlphaFold2, which agrees with previous observations on the limitations of AlphaFold2. In addition, a region relevant to this study involving residues from 151 to 160 of chain F was folded incorrectly by AlphaFold2. Hence, we decided to directly mutate our target residues on the crystal structure of ABE8e, although this assumes there is no conformational or folding change for our mutants.

Using the command line in ChimeraX, we visualized, mutated and analyzed interactions of target residues. We also analyzed interactions of nucleotides 5 to 8 in the gRNA with residues of ABE8e. We sought for hydrogen bonds, nonpolar (van der Waals) interactions between carbon atoms in the gRNA and protein at a maximum distance of 3.8 Å, and cationic interactions between nitrogen atoms within 5 Å of an aromatic carbon involving our target residues and nucleotides.

Editing of iPS cells. Plasmid nucleofection was performed using the Neon Electroporation System (Thermo Fisher) 10 µl kit. A total of 200,000 cells were resuspended in ~10 µl of buffer R and mixed with 200 ng of gRNA vector and 200 µg of BE. Nucleofection was performed using the following parameters: voltage, 1,400 V; width, 20 ms; pulses, two pulses. After nucleofection, cells were plated in 12-well plates with 400 µl of StemFlex medium (Gibco, cat. no. A3349401) without antibiotic and 1:100 dilution of RevitaCell Supplement (Gibco, cat. no. A26445-01). After 24 h, the medium was replaced and editing was evaluated after 48 h by NGS (Quintara Biosciences). gRNAs were generated for this study.

Statistics and reproducibility. No statistical methods were used to predetermine sample size. Sample sizes were selected based on standards from previous genome editing studies and were sufficient to ensure reproducibility. For cell-based and sequencing experiments, three to six independent replicates were performed, which reliably captured biological variability and produced consistent effect sizes across experiments.

All experiments were performed independently at least three times, unless otherwise stated. Statistical analyses and plots were generated using GraphPad Prism v.10 (GraphPad Software LLC). Two-tailed, unpaired Student's *t*-tests, one-way and two-way analysis of variance (ANOVA) were used to evaluate differences between groups, with $P < 0.05$ considered statistically significant. In cases where adding P values directly to the figures would compromise visual clarity, the corresponding statistical comparisons are provided in Supplementary Table 10.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All oligonucleotides, sequences of the new ABE variants, PANCE sequencing evaluation and gRNAs used in the PANCE selection are included in the Supplementary Data. Statistics and plots were generated using PRISM v.10 (Graphpad Software LLC). The ABE variant expression plasmids will be deposited on Addgene: V28C (ABEx1), L34W (ABEx2), M151E (ABEx3) and V28C-M151E (ABEx4). NGS data have been deposited at the NCBI Sequence Read Archive database under accession PRJNA1112370 (ref. 38).

References

34. Zeng, H. et al. A split and inducible adenine base editor for precise in vivo base editing. *Nat. Commun.* **14**, 5573 (2023).
35. Yang, L. et al. Engineering APOBEC3A deaminase for highly accurate and efficient base editing. *Nat. Chem. Biol.* **20**, 1176–1187 (2024).
36. Badran, A. H. et al. Continuous evolution of *Bacillus thuringiensis* toxins overcomes insect resistance. *Nature* **533**, 58–63 (2016).
37. Badran, A. H. & Liu, D. R. Development of potent in vivo mutagenesis plasmids with broad mutational spectra. *Nat. Commun.* **6**, 8425 (2015).
38. Perrotta, R. Enhancing base editing precision via directed evolution. *Sequence Read Archive* https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA1112370 (2025).

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Author contributions

R.M.P. and S.V. conceptualized the project, led the analysis and wrote the manuscript with input from all authors. G.M.C. supervised research. R.M.P. performed wet laboratory experiments with help of all authors. R.M.P. designed, cloned and tested all libraries. S.V. designed the directed evolution systems and libraries. R.F. led the bioinformatic analysis. M.M. led the ML protein design. A.M. performed path_var library experiments in HEK293T cells. A.C.-P. performed the in silico mutagenesis and modeling. L.M.R. developed the evaluation score of agRNA clones. L.M.R., A.-T.M. and L.S.L. contributed to the execution of wet laboratory experiments.

Competing interests

R.M.P., S.V., M.M. and G.M.C. filed a patent application related to the work presented in this study. G.C. is a cofounder of Editas Medicine and has other financial interests listed at: https://arep.med.harvard.edu/gmc/tech.html. The other authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Ramiro M. Perrotta or George M. Church.

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